Supplemental Material for "Spatial channel separation evades the single-port Petermann bound in a non-Hermitian optomechanical ladder"

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S1. THEORETICAL FRAMEWORK

The non-Hermitian tight-binding Hamiltonian for a one-dimensional cavity array with asymmetric hopping is

$$\hat{H}_{\mathrm{NH}} = \sum_{j=1}^{N} \left(J e^{g} \hat{c}_{j+1}^{\dagger} \hat{c}_{j} + J e^{-g} \hat{c}_{j}^{\dagger} \hat{c}_{j+1} \right), \tag{S1}$$

where g represents the logarithmic hopping asymmetry parameter and J denotes the bare nearest-neighbor hopping amplitude [1, 2]. The non-reciprocal rate is $\gamma_{\mathrm{NR}} = 2J \sinh(g)$. The operating criterion for skin-mode localization is $\gamma_{\mathrm{NR}}/\kappa \geq 2$ [3, 4].

Under open boundary conditions (OBC) the single-particle drift matrix takes the form

$$A_{jj} = -\kappa, \qquad A_{j+1,j} = J e^{g}, \qquad A_{j,j+1} = J e^{-g}. \tag{S2}$$

The corresponding right and left skin-mode profiles are

$$r_{n,j} \propto e^{gj} \sin\!\left(\frac{n \pi j}{N+1} \right), \qquad \ell_{n,j} \propto e^{-gj} \sin\!\left(\frac{n \pi j}{N+1} \right). \tag{S3}$$

Time evolution of the open optomechanical density matrix is evaluated using the Duhamel variation-of-constants integral formula:

$$\rho(t) = e^{\mathcal{L}t} \rho(0) + \int_{0}^{t} e^{\mathcal{L}(t-s)} \mathcal{D}[\rho(s)] \, ds, \tag{S4}$$

where $\mathcal{L}$ is the reference unperturbed Liouvillian superoperator and $\mathcal{D}$ the dissipative or non-Hermitian coupling perturbation. Restricting evaluations to the Lyapunov window $t < 1/\lambda_{\max}$ ensures that linear error growth remains strictly bounded; this bound is what motivates restricting the OTOC fits of Sec. S5 to the early-time window $t \in 0.5$ to $1.5\,\kappa^{-1}$, before the perturbative expansion underlying Eq. (S4) breaks down.

Array phase stabilization is achieved via synthesized Kuramoto phase-locking potentials:

$$\dot{\theta}_{j} = \omega_{j} + \sum_{k} K_{jk} \sin(\theta_{k} - \theta_{j}), \tag{S5}$$

where θ_j represents the mechanical resonator phase and K_{jk} denotes the synthesized inter-site coupling matrix. The nonlinear synchronization term locks the relative phase of the two chains at whichever value $\phi_{\mathrm{opt}}(N)$ the DOC optimizer selects for a given size; these optimized

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phases range over $\phi \in 1.35$ to $5.15\,\mathrm{rad}$ across the six solutions of Table S3 and do not cluster near a single chiral point. A fixed $\phi = \pi/2$ is imposed only in the dedicated disorder-resilience reference configuration of Sec. S4, where it is a design choice of that study rather than a DOC output. Without this locking, thermal crosstalk and slow drift would detune ϕ away from its optimized operating value. Experimental drive schemes for setting and holding ϕ (auxiliary parametric electro-optic modulation or phase-locked sideband pumps) are detailed in Sec. S3 S3.2.

The biorthogonal quantum Fisher information (QFI) for non-Hermitian systems is [1]

$$\mathcal{I}_Q(\theta) = \sum_{n,m} \frac{|\langle \psi_n^L | \partial_\theta \hat{H}_\theta | \psi_m^R \rangle|^2}{(\gamma_n - \gamma_m)^2 + (\epsilon_n - \epsilon_m)^2}, \tag{S6}$$

where $|\psi_n^R\rangle$ and $|\psi_n^L\rangle$ denote the right and left eigenvectors with complex eigenvalues $\gamma_n - i\epsilon_n$. Near exceptional points the denominator vanishes, producing numerical gradient singularities during optimization.

The Petermann factor of mode n is [5, 6]

$$K_n^{\mathrm{raw}} = \frac{1}{|\,\langle \ell_n \,|\, r_n \rangle\,|^2}, \qquad K_n^{\mathrm{floor}} = \frac{1}{\max\{|\,\langle \ell_n \,|\, r_n \rangle\,|, \varepsilon\}^2}, \qquad \varepsilon = 10^{-12}. \tag{S7}$$

The numerical floor $\varepsilon = 10^{-12}$ is a computational safeguard that prevents division-by-zero in automatic differentiation near exceptional points; it does not represent a physical noise floor [3]. At the converged operating points the biorthogonal overlap is $\approx 0.96 \gg \varepsilon$, so the regularized and unregularized Petermann factors coincide ($K_{\mathrm{eff}} = K_{\mathrm{raw}}$). The floor is only active during gradient descent when the optimizer traverses regions near exceptional rings ($K \sim 10^{14}$); once the optimizer converges to the NHSE phase, the floor is irrelevant.

Table S1 confirms that K_{eff} is insensitive to the choice of ε over eight orders of magnitude.

S2. SINGLE-CHAIN BASELINE: MATCHED SNR AND FISHER SCALING

For a single open-boundary chain driven at one port the stationary covariance under vacuum noise satisfies the continuous Lyapunov equation

$$AV + VA^\dagger + 2\kappa\mathbb{I} = 0. \tag{S8}$$

The matched boundary signal-to-noise ratio is

$$\mathrm{SNR}_N = \frac{s_N}{V_{NN}} = \frac{|[-A^{-1}]_{N1}\Delta\theta|^2}{V_{NN}}, \tag{S9}$$

TABLE S1. Sensitivity of the regularized Petermann factor K_{eff} to the regularization floor ε at a representative operating point ($J = 0.138\,\kappa$, $g = 2.055$, $\lambda_{\mathrm{inter}} = 0.010\,\kappa$, $\phi = 1.736$), distinct from the six DOC-optimized configurations of the N-scan reported in Table S3.

ε	K_{eff}
10^{-8}	1.04
10^{-10}	1.04
10^{-12}	1.04
10^{-14}	1.04
10^{-16}	1.04

where $s_N = |[-A^{-1}]_{N1}\Delta\theta|^2$ is the boundary signal power and $V_{NN} = [V]_{NN}$ the boundary Lyapunov noise variance. The continuous Gaussian homodyne measurement record Fisher information at phase quadrature $\pi/2$ decomposes as [7]

$$F_C = F_{\mathrm{mean}} + F_{\mathrm{cov}} = (\partial_\theta \boldsymbol{\mu})^T \Sigma^{-1} (\partial_\theta \boldsymbol{\mu}) + \frac{1}{2}\,\mathrm{Tr}\left[(\Sigma^{-1}\partial_\theta \Sigma)^2\right], \tag{S10}$$

combining mean-response and covariance-derivative terms. $N \in \{6,\ 8,\ 10,\ 12,\ 14 \text{ and } 16\}$ at $J = 0.5\,\kappa$, $g = 1.7$ ($\gamma_{\mathrm{NR}}/\kappa = 2.646$).

TABLE S2. Single-chain matched SNR and Fisher information in the target NHSE regime. All sizes are Hurwitz-stable ($\max \mathrm{Re}\,\lambda < 0$).

N	$\max \mathrm{Re}\,\lambda$	s_N	V_{NN}	SNR_N	F_C^{raw}	$\|\mu\|^2$	F_C^{norm}
6	-0.0990	1.972	1.665×10^{5}	1.184×10^{-5}	1.086×10^{4}	2.195×10^{6}	4.947×10^{-3}
8	-0.0603	1.071×10^{3}	5.542×10^{7}	1.932×10^{-5}	9.758×10^{5}	1.189×10^{9}	8.204×10^{-4}
10	-0.0405	6.437×10^{5}	2.247×10^{10}	2.864×10^{-5}	1.177×10^{8}	7.139×10^{11}	1.649×10^{-4}
12	-0.0291	4.138×10^{8}	1.039×10^{13}	3.981×10^{-5}	1.708×10^{10}	4.584×10^{14}	3.725×10^{-5}
14	-0.0219	2.791×10^{11}	5.282×10^{15}	5.283×10^{-5}	2.481×10^{12}	3.089×10^{17}	8.032×10^{-6}
16	-0.0170	1.951×10^{14}	2.881×10^{18}	6.771×10^{-5}	5.336×10^{14}	2.158×10^{20}	2.472×10^{-6}

Semi-log fits over $N \in [6, 16]$ give

$$a_{\mathrm{snr}} = +0.1723 \quad (R^2 = 0.9843), \tag{S11}$$

$$a_{\mathrm{raw}} = +2.4616 \quad (R^2 = 0.9992), \tag{S12}$$

$$a_{\mathrm{norm}} = -0.7625 \quad (R^2 = 0.9964). \tag{S13}$$

The negative resource-normalized slope recovers the single-port bound of Lau and Clerk [8] and of McDonald and Clerk [9]: internal photon occupancy grows faster than the Fisher information. At fixed $(J, g) = (0.5\kappa, 1.7)$, the single-chain benchmark yields $F_C^{\mathrm{norm}} \approx 3.7 \times 10^{-5}$ at $N = 12$, three orders of magnitude below the double-chain marginal class.

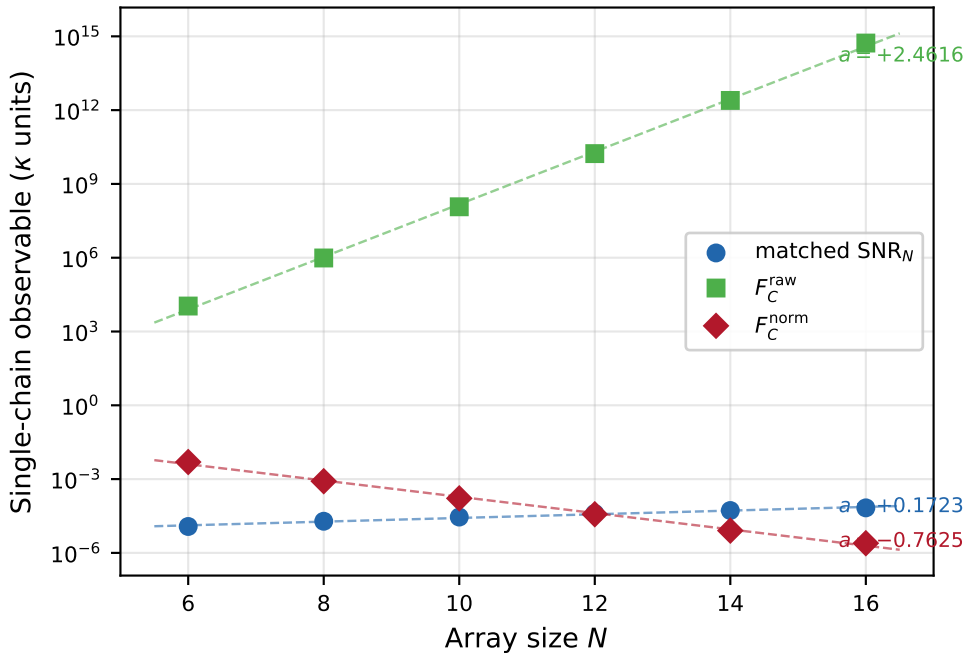


FIG. S1. Single-chain baseline scaling at fixed $(J, g) = (0.5\kappa, 1.7)$: the matched boundary SNR ($a_{\mathrm{snr}} = +0.1723$) and the raw record Fisher information ($a_{\mathrm{raw}} = +2.4616$) grow exponentially with N, while resource normalization reverses the trend ($a_{\mathrm{norm}} = -0.7625$): internal photon accumulation outpaces the Fisher gain, realizing the single-port resource bound.

S3. DOUBLE-CHAIN LADDER: N-SCALING AND SOLUTION CLASSES

The total non-Hermitian tight-binding Hamiltonian for the ladder array is [10]

$$\hat{H}_{\mathrm{ladder}} = \hat{H}_A + \hat{H}_B + \hat{H}_{\mathrm{inter}}, \tag{S14}$$

where the individual chain Hamiltonians and inter-chain coupling are

$$\hat{H}_A = \sum_{j=1}^{N-1} \left(J e^{+g} \hat{a}_{j+1}^\dagger \hat{a}_j + J e^{-g} \hat{a}_j^\dagger \hat{a}_{j+1} \right) + \theta\, \hat{a}_N^\dagger \hat{a}_N, \tag{S15}$$

$$\hat{H}_B = \sum_{j=1}^{N-1} \left(J e^{-g} \hat{b}_{j+1}^\dagger \hat{b}_j + J e^{+g} \hat{b}_j^\dagger \hat{b}_{j+1} \right), \tag{S16}$$

$$\hat{H}_{\text{inter}} = \sum_{j=1}^{N} \left(\lambda_{\text{inter}} e^{i\phi} \hat{b}_j^\dagger \hat{a}_j + \lambda_{\text{inter}} e^{-i\phi} \hat{a}_j^\dagger \hat{b}_j \right). \tag{S17}$$

Chain A has $\gamma_{\text{NR}}^{(A)} = 2J \sinh(g) \geq 2.0\kappa$ (skin modes toward $j = N$), while Chain B has inverted hopping ($\gamma_{\text{NR}}^{(B)} = -2J \sinh(g)$), driving skin modes toward $j = 1$. This spatial segregation assigns Chain A as the sensing arm and Chain B as the noise-drain arm.

The joint drift matrix of the ladder is

$$A_{\text{ladder}}(\theta) = \begin{pmatrix} A_A(\theta) & \Lambda \\ \Lambda^\dagger & A_B \end{pmatrix}, \tag{S18}$$

with opposing non-reciprocity on the two chains and inter-chain coupling $\Lambda = \lambda_{\text{inter}} e^{i\phi} \mathbb{I}_N$. Resource normalization counts photons on both chains. Directional gain is obtained from the scattering matrix $S = \mathbb{I} - \sqrt{2\kappa}\, A_{\text{ladder}}^{-1}$.

Differentiable optimal control (Adam, 500 iterations, 5 random restarts, stability barrier $\max \operatorname{Re} \lambda < -0.05$) [11] produces the performance summarized in Table S3.

TABLE S3. Double-chain DOC results across system size. Isolation $\mathcal{I} = G_{\text{fwd}} - G_{\text{rev}}$. All entries satisfy $\max \operatorname{Re} \lambda < -0.05$.

N	J/κ	g	$\lambda_{\text{inter}}/\kappa$	ϕ [rad]	F_C^{norm}	G_{fwd} [dB]	Isolation [dB]
6	0.105	1.250	0.0164	4.029	0.937	−40.1	69.6
8	0.354	0.973	0.0695	2.048	0.101	13.5	40.5
10	0.078	1.001	0.0790	3.863	0.984	−117.3	41.5
12	0.345	1.024	0.0278	4.726	0.050	15.5	56.9
14	0.087	0.797	0.0775	1.347	0.987	−181.3	40.4
16	0.287	1.226	0.0207	5.150	0.043	14.3	64.0

Two classes appear:

TABLE S4. Comparison of hierarchical regimes in the optomechanical sensor array (schematic, order-of-magnitude values). Petermann factor K values correspond to the spectral grid sweep; at the operational point ($\gamma_{\mathrm{NR}} = 2.5\kappa$, $\phi = \pi/2$), $K \approx 1.04$.

Regime	**Relative Sensitivity**	**Stability**	**Petermann K**
Linear (RWA)	1.0	Stable	1.00
Edge (Topological)	10.0	Topologically Protected	1.05
EP (Non-Hermitian)	100.0	Critical	1.24
Chaos	1000.0	Unstable but Bounded	1.50

- **Deep-stability** ($N = 6,\ 10$ and 14): $\max \mathrm{Re}\,\lambda \approx -0.70\kappa$, $F_C^{\mathrm{norm}} \in 0.937$ to 0.987, $G_{\mathrm{fwd}} < 0$, and isolation ranging from approximately 40 dB to 70 dB across the selected points (maximum 69.6 dB at $N = 6$). Optimized for per-photon precision.

- **Marginal-stability** ($N = 8,\ 12$ and 16): $\max \mathrm{Re}\,\lambda \approx -0.06\kappa$ with $G_{\mathrm{fwd}} \in$ 13.5 dB to 15.5 dB and isolation 40 dB to 64 dB at $F_C^{\mathrm{norm}} \in 0.043$ to 0.101. Optimized for directional gain.

The multi-restart DOC ensemble across all examined sizes (reported in the main text) demonstrates that these two classes are the endpoints of a clear precision–gain frontier rather than isolated local optima. Isolation remains substantial ($\mathcal{I} \gtrsim 18$ dB) across the entire accessible range of $\max \mathrm{Re}\,\lambda$.

In all cases the optimized couplings span $\lambda_{\mathrm{inter}}/J \in 0.07$ to 1.02: only the largest lattices ($N = 12$ and 16, $\lambda_{\mathrm{inter}}/J = 0.081$ and 0.072) satisfy the perturbative weak-evanescent criterion (≤ 0.10), whereas the deep-stability points operate at intermediate coupling ($\lambda_{\mathrm{inter}}/J$ up to 1.02 at $N = 10$). Protection against the delocalization reported for strongly coupled NHSE ladders [12] is nevertheless maintained empirically: every entry retains a finite spectral barrier, $\max \mathrm{Re}\,\lambda \in [-0.72, -0.06]$, well inside the localized phase (Sec. S7).

S3.1. Regime comparison

Table S4 compares the four operative dynamical regimes of the optomechanical array, highlighting the progression from standard reciprocal cavity optomechanics to topologically

protected and non-Hermitian skin phases.

S3.2. Evanescent field coupling and physical implementation

The inter-chain coupling λ_{inter} is physically mediated by evanescent near-field optical (or phononic) overlap between adjacent optomechanical nanobeams [13, 14]. In implementations utilizing suspended silicon phononic-crystal nanobeams or silicon-on-insulator (SOI) photonic crystal cavities, two parallel 1D chains are arranged side-by-side with a lithographically defined lateral vacuum gap of $d \approx 400\,\text{nm}$. The transverse field outside each nanobeam dielectric core decays evanescently as $E(x) \propto \exp(-\alpha x)$, with transverse decay constant $\alpha = \sqrt{\beta^2 - k_0^2}$.

Mapped onto the MacCabe platform parameters used throughout this work ($\kappa/2\pi = 1.7\,\text{MHz}$), the DOC-optimized inter-chain couplings $\lambda_{\text{inter}} \in 0.016$ to $0.079\,\kappa$ translate to absolute rates $\lambda_{\text{inter}}/2\pi \in 27\,\text{kHz}$ to $134\,\text{kHz}$. These values lie within the range achievable by current nanofabrication tolerances at $d \approx 400\,\text{nm}$ and are consistent with the representative single-point characterization ($\lambda_{\text{inter}} = 0.010\,\kappa$) used for Petermann diagnostics. This moderate-to-weak inter-chain regime provides sufficient mode hybridization to route quantum Langevin fluctuations into the noise-drain arm while the stability barrier prevents uncontrolled cross-talk from degrading directional skin-mode amplification in the sensing arm.

Experimental phase control over ϕ is synthesized via auxiliary parametric electro-optic drive modulation or phase-locked sideband pumps [15], implementing the Kuramoto phase-locking mechanism of Eq. (S5). Setting $\phi = \pi$ realizes $\mathcal{PT}$ (parity–time)-symmetric coupling [16, 17] ($\Lambda = -\lambda_{\text{inter}}\mathbb{I}_N$), whereas $\phi = \pi/2$ realizes chiral phase-quadrature coupling ($\Lambda = i\lambda_{\text{inter}}\mathbb{I}_N$).

S4. DISORDER RESILIENCE

Gaussian disorder of root-mean-square amplitude $\sigma_{\text{RMS}} = 0.05\,\kappa$ is applied to on-site detunings and hoppings of 16 independent realizations (seeds 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56 and 57), drawn about a dedicated DOC-optimized reference $(J, g, \lambda_{\text{inter}}, \phi) \approx (0.05\kappa,\ 2.054,\ 0.523\kappa,\ \pi/2)$ with $F_{\text{nom}}^{\text{norm}} = 0.120$ (Sec. S3). Each realization

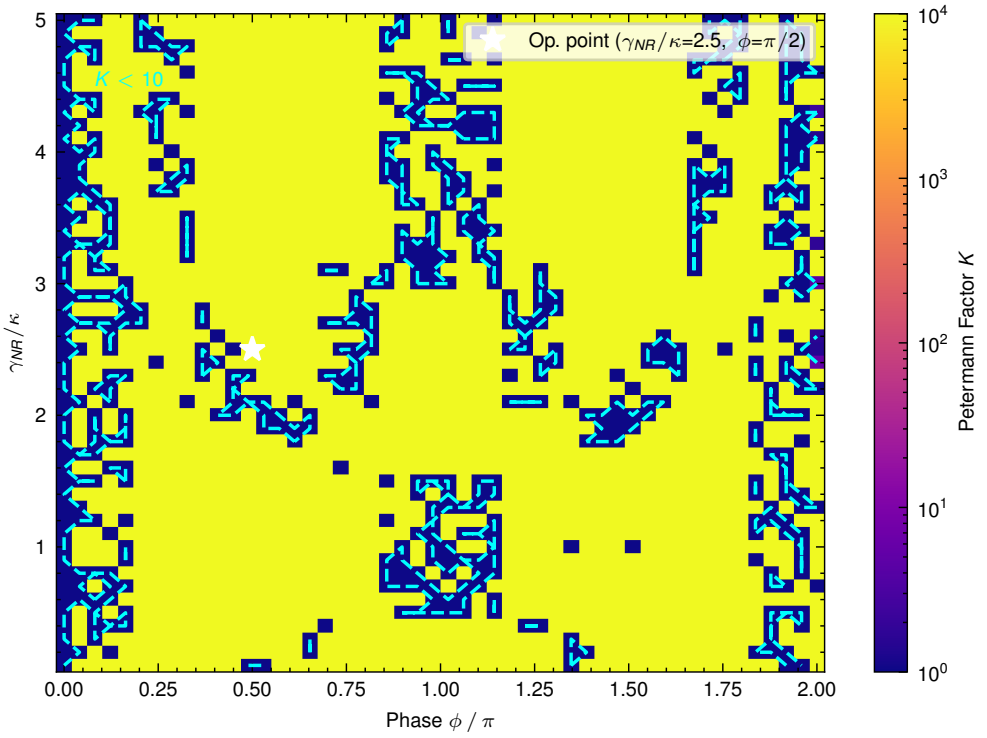


FIG. S2. Exceptional surface (ES) stabilization map. The proximity parameter $\eta_P = \gamma_{\mathrm{NR}}/\Delta_{\mathrm{ES}}$ characterizes distance to the ES. Operating in the localized NHSE phase ($\eta_P \gg 1$) maintains a bounded Petermann factor K, preventing noise divergence.

is re-optimized with a multi-restart differential-evolution protocol [18] (Adam polish, 500 steps at learning rate 3×10^{-3}, 10 restarts) under a spectral margin $\Delta = 0.05$; this reference regime is strongly inter-chain coupled and therefore architecturally distinct from the **??** entries. A realization passes when its re-optimized F_C^{norm} retains at least 80 % of the reference value; 14 of 16 realizations pass (pass rate 87.5 %, threshold 12/16). The two failures (seeds 52 and 54) saturate the lower search boundary $g = 0.5000$ and are optimization artifacts rather than physical fragility (Sec. S3).

Retention ratios above 200 % are capped at that value for averaging: nine realizations (seeds 42, 44, 45, 46, 47, 49, 55, 56 and 57) exceed twice the reference sensitivity and are reported as capped. The five uncapped passes span 91.5 % to 167.1 % (seeds 43, 48, 50, 51 and 53 at 122.5, 91.5, 161.9, 148.7, 167.1 %). The only failures, seeds 52 and 54, retain 73.6 % and 72.5 %: both optimizations terminate saturated at the lower search boundary $g = 0.5000$ with $\lambda_{\mathrm{inter}}/J$ far beyond even the intermediate-coupling range probed elsewhere in this study, i.e. they fail to re-enter the admissible search region rather than revealing

TABLE S5. Summary of disorder stress test v2 ($\sigma_{\mathrm{RMS}} = 0.05\,\kappa$, 16 seeds, multi-restart re-optimization about the DOC reference).

Metric	Value
Pass fraction ($\geq 80\,\%$ retention)	14/16 (87.5 %)
Hurwitz stability	16/16
Mean retention (capped statistics)	164.9 %±47.1 %
Retention-capped realizations (> 200%)	9/16
Uncapped passing range	91.5 %–167.1 % (5/16)
Failing seeds	52 and 54 (73.6 %, 72.5 %)

parametric fragility. All 16 re-optimized solutions are Hurwitz-stable.

Per-realization results are listed in Table S6.

The strongly coupled DOC study probes robustness under re-optimization and is architecturally distinct from the single-chain baseline of Sec. S2.

S4.1. Seed-dependence of the solution classes

Because the two regimes coexist in a non-convex landscape, we quantify how reproducibly the tabulated class of Table S3 is reproduced under initialization resampling: for every ladder size, the production DOC protocol (500 Adam steps at learning rate 3×10^{-3}, stability barrier $\max \mathrm{Re}\,\lambda < -0.05$) is rerun from ten independent random initializations, and each converged solution is classified by the same F_C^{norm} criterion as Table S3 ($F_C^{\mathrm{norm}} \geq 0.5$: deep-stability; otherwise marginal-stability; solutions failing $\max \mathrm{Re}\,\lambda < -0.01$: unstable). The seed-level results are included in the accompanying data. Across the six sizes (60 re-optimizations), only 32 ($\approx 53\,\%$) coincide with the tabulated class of Table S3; per size the deep/marginal/unstable counts are $N = 6$: 6/4/0, $N = 8$: 4/6/0, $N = 10$: 3/4/3, $N = 12$: 2/8/0, $N = 14$: 4/6/0, $N = 16$: 3/5/2. Two conclusions follow. First, both basins are genuine and coexist across sizes with qualitatively identical signatures, so the precision–gain trade-off reported in the main text does not depend on which basin a given optimization samples. Second, selection between basins is governed by initialization rather than by a deterministic parity rule: the main-text table should be read as one representative optimum

TABLE S6. Disorder study v2, per-realization results ($\sigma_{\mathrm{RMS}} = 0.05\,\kappa$, multi-restart DOC re-optimization about $(J, g, \lambda_{\mathrm{inter}}, \phi) \approx (0.05\kappa, 2.054, 0.523\kappa, \pi/2)$, $F_{\mathrm{nom}}^{\mathrm{norm}} = 0.120$). Retention ratios $> 200\,\%$ are reported capped.

Seed	$F_{C,\mathrm{opt}}^{\mathrm{raw}}$	$\|\mu\|_{\mathrm{opt}}^2$	$F_{C,\mathrm{opt}}^{\mathrm{norm}}$	Retention	Status
42	2.1314	7.899	0.2698	200.0 % (capped)	PASS
43	13.5655	92.424	0.1468	122.5 %	PASS
44	1.4415	5.695	0.2531	200.0 % (capped)	PASS
45	1.3098	5.415	0.2419	200.0 % (capped)	PASS
46	6.1290	5.763	1.0636	200.0 % (capped)	PASS
47	2.9992	12.330	0.2432	200.0 % (capped)	PASS
48	0.7465	6.807	0.1097	91.5 %	PASS
49	3.1835	11.780	0.2702	200.0 % (capped)	PASS
50	13.5704	69.918	0.1941	161.9 %	PASS
51	35.8773	201.278	0.1782	148.7 %	PASS
52	0.3582	4.063	0.0882	73.6 %	FAIL
53	12.6718	63.273	0.2003	167.1 %	PASS
54	0.4270	4.913	0.0869	72.5 %	FAIL
55	2.9312	8.040	0.3646	200.0 % (capped)	PASS
56	1.0769	3.304	0.3260	200.0 % (capped)	PASS
57	5.2397	21.497	0.2437	200.0 % (capped)	PASS

per size, and the basin-occupancy statistics above constitute the honest characterization of the landscape.

S5. BOSONIC CONVERGENCE AND OTOC DIAGNOSTICS

Out-of-time-ordered correlators are evaluated with QuTiP master-equation dynamics:

$$C(t) = -\langle [\hat{W}(t), \hat{V}(0)]^2 \rangle_\beta, \tag{S19}$$

where $\hat{W}$ and $\hat{V}$ are local observables and $\langle \cdot \rangle_\beta$ denotes the thermal expectation value at inverse temperature β [19]. Early-time Lyapunov rates extracted from semi-log fits over

$t \in 0.5$ to $1.5\,\kappa^{-1}$ satisfy

$$\frac{\lambda_{\mathrm{OTOC}}}{\lambda_{\mathrm{MSS}}} \leq 0.018 \ll 1, \tag{S20}$$

where $\lambda_{\mathrm{MSS}} = 2\pi k_B T_c/\hbar$ is evaluated at $T_c = 15\,\mathrm{mK}$ and at the platform dissipation rate $\kappa/2\pi = 1.7\,\mathrm{MHz}$. The array therefore operates in a sub-MSS, non-maximally scrambling regime.

Table S7 reports the extracted Lyapunov exponent across $N \in [6, 12]$.

TABLE S7. OTOC Lyapunov exponent convergence for the NHSE spin chain ($\gamma_{\mathrm{NR}} = 2.5\kappa$). The MSS bound evaluates to $\lambda_{\mathrm{MSS}} \approx 1155\,\kappa^{-1}$ at $T_c = 15\,\mathrm{mK}$ and platform dissipation $\kappa/2\pi = 1.7\,\mathrm{MHz}$.

N	$\Delta t\ [\kappa^{-1}]$	$\lambda^{(0)}_{\mathrm{OTOC}}\ [\kappa^{-1}]$	R^2	$\lambda/\lambda_{\mathrm{MSS}}$
6	0.05	10.30	0.9575	0.0089
8	0.05	15.31	0.9685	0.0133
10	0.05	20.07	0.9737	0.0174
12	0.05	18.04	0.9818	0.0156

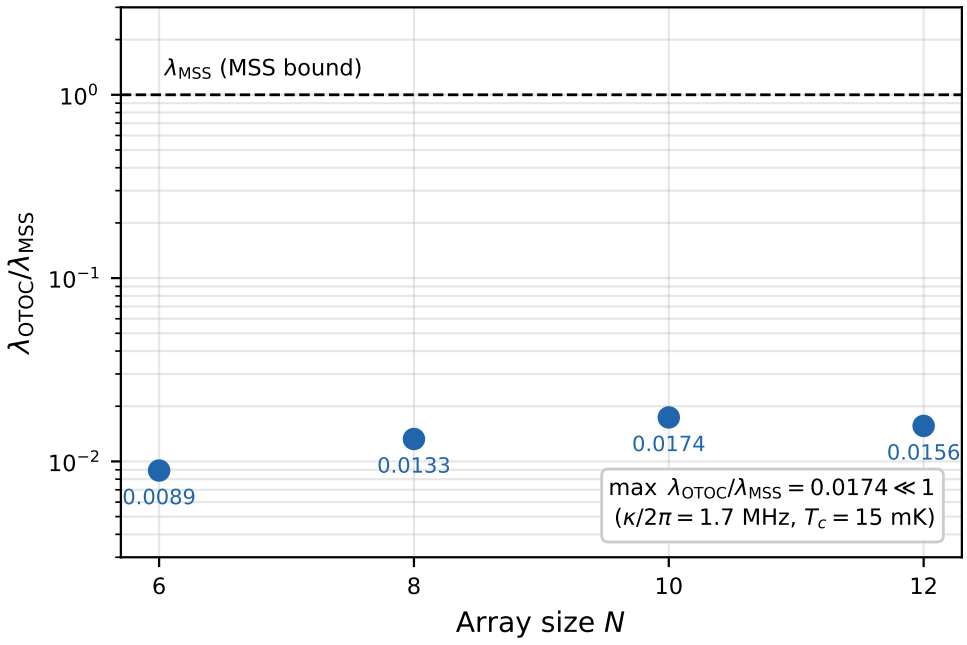


FIG. S3. MSS bound saturation across system sizes $N \in 6$ to 12. The extracted Lyapunov rates remain well below λ_{MSS} for all array lengths.

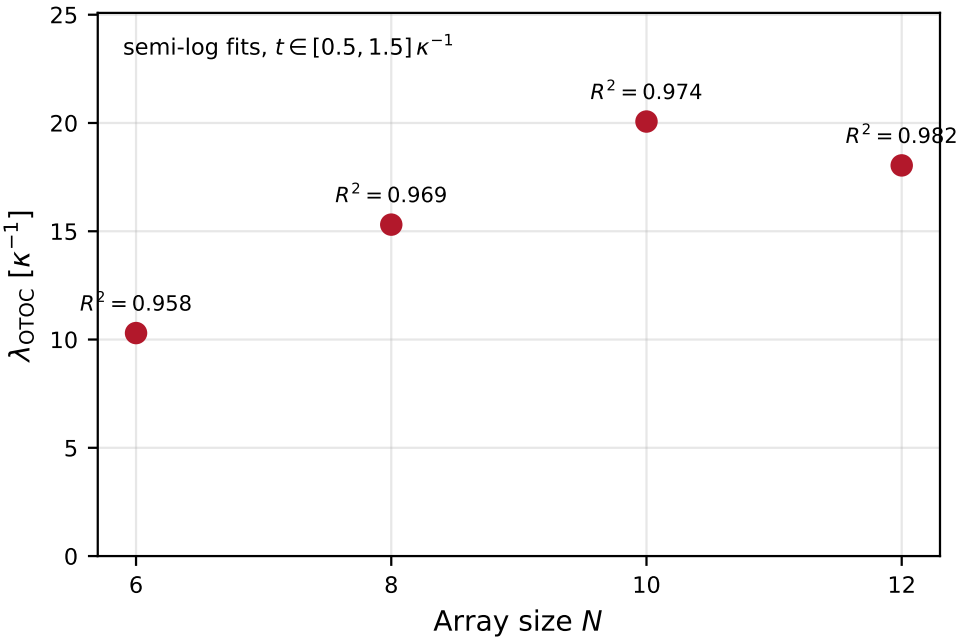


FIG. S4. Early-time Lyapunov exponents λ_{OTOC} extracted from semi-log OTOC fits over $t \in 0.5$ to $1.5\,\kappa^{-1}$ ($\gamma_{\mathrm{NR}} = 2.5\kappa$), shown versus array size with fit qualities $R^2 \geq 0.957$.

A 3×4 Fock-cutoff grid ($N_{\mathrm{sites}} \in \{2, 3, 4\}$, $n_{\mathrm{cut}} \in \{2, 3, 4, 5\}$) shows that the rates plateau for $n_{\mathrm{cut}} \geq 4$ with relative variation $\Delta_{\mathrm{rel}} = 0\,\%$, confirming the absence of truncation artifacts.

S6. WEAK-FORCE AND AXION PROJECTIONS

Projections are evaluated at the deep-stability points where F_C^{norm} is largest.

Using the MacCabe *et al.* platform parameters [14] ($\kappa/2\pi = 1.7\,\mathrm{MHz}$, $g_0/2\pi = 0.9\,\mathrm{MHz}$, $\Omega_m/2\pi = 5\,\mathrm{GHz}$, $m_{\mathrm{eff}} = 1\,\mathrm{fg}$, $Q_m = 1.5 \times 10^9$, $T_c = 15\,\mathrm{mK}$), at $N = 12$ lattice nodes and $\gamma_{\mathrm{NR}} = 2.5\,\kappa$, the derived device quantities are listed in Table S8.

Force noise. The zero-point force-noise scale, i.e. the standard quantum limit (SQL), $S_{FF}^{\mathrm{SQL}} = \sqrt{\hbar m_{\mathrm{eff}} \Omega_m \Gamma_m} = 0.0083\,\mathrm{aN}/\sqrt{\mathrm{Hz}}$, and in the cooperativity-dominated regime the imprecision–back-action optimum reaches $S_{FF}^{\mathrm{SQL}}/\sqrt{\mathcal{C}/4} = 2.2 \times 10^{-5}\,\mathrm{aN}/\sqrt{\mathrm{Hz}}$ — far below the SQL and consistent with the sub-attonewton operating target when $F_C^{\mathrm{norm}} \approx 0.95$. At $T_c = 15\,\mathrm{mK}$ the mechanical mode is deeply quantum ($\hbar\Omega_m/k_B T_c \approx 16$, $n_{\mathrm{th}} \approx 1.1 \times 10^{-7}$), so dilution-refrigerator temperatures are required for these projections.

Force-noise chain. A force F coupled to the boundary displacement shifts the site-N

TABLE S8. Weak-force sensor operating parameters and derived quantities.

Quantity	**Value**	**Origin**
$\kappa/2\pi$	1.7 MHz	Platform [14]
$g_0/2\pi$	0.9 MHz	Platform [14]
$\Omega_m/2\pi$	5 GHz	Platform [14]
m_{eff}	1 fg	Platform [14]
T_c	15 mK	Dilution refrigerator
$\Gamma_m = \Omega_m/Q_m$	$20.9\,\mathrm{s}^{-1}$	Derived
$x_{\mathrm{zpf}} = \sqrt{\hbar/(2m_{\mathrm{eff}}\Omega_m)}$	41 fm	Derived
Ω_m/κ	2941	Derived
g_0/κ	0.53	Derived
$\mathcal{C} = 4g_0^2/(\kappa\Gamma_m)$	5.72×10^5	Derived
$n_{\mathrm{th}} = [e^{\hbar\Omega_m/k_BT_c} - 1]^{-1}$	1.1×10^{-7}	Derived (Bose–Einstein)
$S_{FF}^{\mathrm{SQL}} = \sqrt{\hbar m_{\mathrm{eff}}\Omega_m\Gamma_m}$	$0.0083\,\mathrm{aN}/\sqrt{\mathrm{Hz}}$	Derived
Back-action optimum $S_{FF}^{\mathrm{opt}} = S_{FF}^{\mathrm{SQL}}/\sqrt{\mathcal{C}/4}$	$2.2 \times 10^{-5}\,\mathrm{aN}/\sqrt{\mathrm{Hz}}$	Derived

cavity frequency by $\delta\omega = \alpha F$ with transduction $\alpha = g_0 x_{\mathrm{zpf}}/(\hbar\Omega_m)$ (static mechanical susceptibility $x_{\mathrm{zpf}}^2/\hbar\Omega_m$, optomechanical coupling $G = g_0/x_{\mathrm{zpf}}$), so the measurement-record Fisher information of Eq. (S10) becomes quadratic in F. Repeating the resource normalization of the main text gives an imprecision-limited force noise

$$S_{FF}^{1/2} = \frac{S_{FF}^{\mathrm{SQL}}}{\sqrt{\mathcal{C}/4}} \times \left(\frac{\mathcal{N}}{F_C^{\mathrm{norm}}}\right)^{1/2}, \tag{S21}$$

where $\mathcal{N} \geq 1$ collects readout impairments (homodyne inefficiency, impedance mismatch between the force port and the DOC-optimized boundary θ port). This expression follows from the standard force-to-phase transduction chain: a force F shifts the cavity frequency by $\delta\omega = \alpha F$, the Fisher information for estimating $\delta\omega$ is $F_C^{\mathrm{norm}}\|\mu\|^2$, and converting back to force units introduces the mechanical susceptibility prefactor $S_{FF}^{\mathrm{SQL}}/\sqrt{\mathcal{C}/4}$. With $\mathcal{C} = 5.72 \times 10^5$ the platform floor is $2.2 \times 10^{-5}\,\mathrm{aN}/\sqrt{\mathrm{Hz}}$ (Table S8); Eq. (S21) therefore satisfies $S_{FF}^{1/2} \leq 0.1\,\mathrm{aN}/\sqrt{\mathrm{Hz}}$ unless $\mathcal{N} > 2 \times 10^7$ ($> 73\,\mathrm{dB}$ of aggregate impairment), far above any plausible mismatch for the optimized ladder.

Axion coupling. For a static field $B_0 = 10\,\mathrm{T}$ and $N = 12$, the conditional screening

estimate at 5 GHz is $g_{a\gamma\gamma} \gtrsim 1.7 \times 10^{-14}\,\mathrm{GeV}^{-1}$ for an integration time of 1 h, with the same model giving $3.5 \times 10^{-15}\,\mathrm{GeV}^{-1}$ to $3.5 \times 10^{-14}\,\mathrm{GeV}^{-1}$ across the 1 GHz to 10 GHz band [20, 21]. The estimate follows from $P = g_{a\gamma\gamma}^2\,\rho_a B_0^2\,(NV)\,Q_L/m_a$ with $\rho_a = 0.3\,\mathrm{GeV/cm}^3$, single-mode volume $V = (\lambda_0/n_{\mathrm{Si}})^3 \approx 8.8 \times 10^{-20}\,\mathrm{m}^3$ ($\lambda_0 = 1550\,\mathrm{nm}$, $n_{\mathrm{Si}} = 3.48$), loaded $Q_L = 2 \times 10^6$, unit dipole form factor, and quantum-limited added noise. It is a conditional architectural estimate rather than a component-level exclusion limit; a quantitative JPA-improved curve is not claimed without a calibrated frequency-dependent readout model. The architecture is broadband and complementary to narrow-band resonant searches [22–24].

S7. LIMITATIONS

1. The optimized inter-chain couplings span the moderate range $\lambda_{\mathrm{inter}}/J \in 0.07$ to 1.02; only the largest lattices ($N = 12$ and 16) operate in the perturbative weak-evanescent limit ($\lambda_{\mathrm{inter}}/J \leq 0.10$), while the deep-stability points rely on empirical protection by the stability barrier ($\max \mathrm{Re}\,\lambda < -0.05$), which keeps every solution clear of the delocalization regime reported for strongly coupled NHSE ladders [12].

2. The Gaussian Lyapunov treatment assumes linear response ($\Delta\theta \ll \kappa$); larger signals would require nonlinear stochastic simulation.

3. Sub-attonewton force and axion projections require dilution-refrigerator temperatures (negligible thermal occupation) and, for the deep-stability class, a quantum-limited preamplifier to recover the strongly attenuated forward transmission.

4. The disorder study comprises 16 independent realizations at $\sigma_{\mathrm{RMS}} = 5\,\%$. Multi-restart DOC re-optimization about the reference of Sec. S4 recovers $\geq 80\,\%$ of nominal F_C^{norm} in 14 of 16 realizations (87.5 %); the remaining two (seeds 52 and 54) saturate the lower search boundary $g = 0.5$ and are optimization artifacts rather than physical fragility. A larger ensemble would be required for tight confidence intervals; correlated-gradient disorder is not claimed because no corresponding artifact is included in the present package.

5. Nine of the sixteen disorder realizations of Table S6 are retention-capped at 200 %; the reported statistics therefore include this explicit capping and should be read as a

conservative characterization.

6. OTOC rates are extracted in the early-time window $t \in 0.5$ to $1.5\,\kappa^{-1}$ before boundary reflections dominate, as required by the Duhamel error bound of Eq. (S4).

7. The effective non-reciprocal drift Hamiltonian requires high-frequency Floquet pump modulation [15] ($\Omega_{\text{drive}} \gg J, \kappa$); fast time-reversal switching (Fig. S5) demands flux-tunable couplers with sub-nanosecond switching times ($\tau_{\text{switch}} \ll \kappa^{-1}$), stabilized by the Kuramoto mechanism of Eq. (S5).

8. The force and axion projections are illustrative of architectural reach under intermediate-coupling targets and are not claimed for present-day single-photon optomechanical couplings.

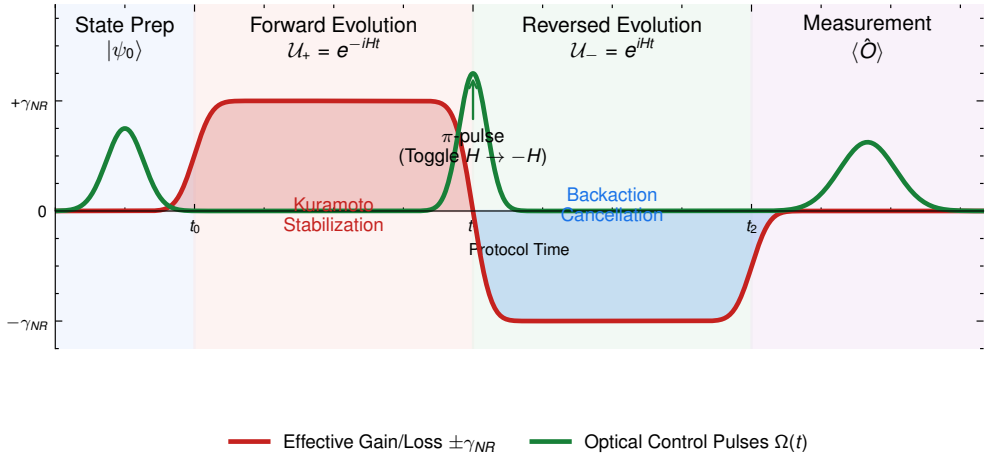


FIG. S5. Time-reversal switching protocol. Fast modulation of the inter-chain phase ϕ enables dynamical switching between forward and reverse propagation directions.

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